

Jan 29, 17:30

This lec: some semantics and syntax of CTL

form vs original artifact



model

∴ most systems are complex

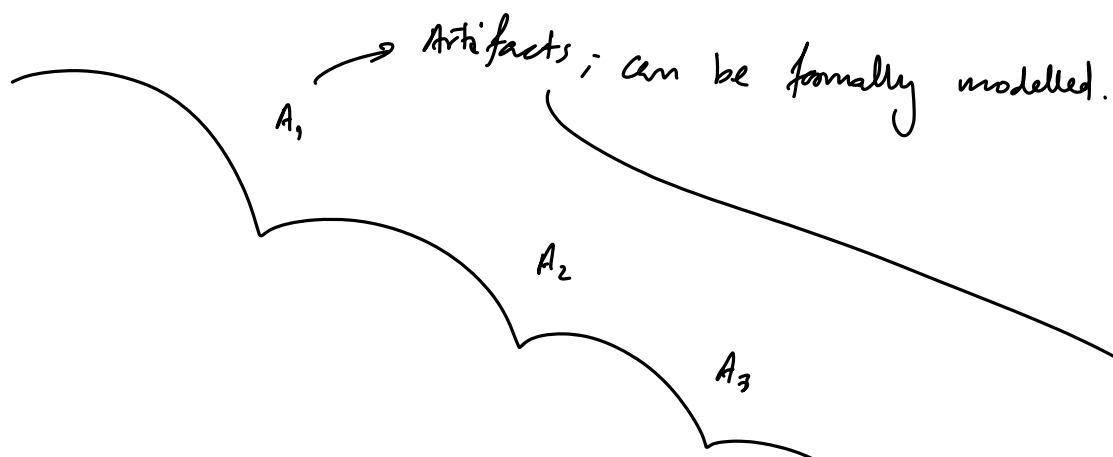
formal methods → only increase correctness, not guarantee correctness

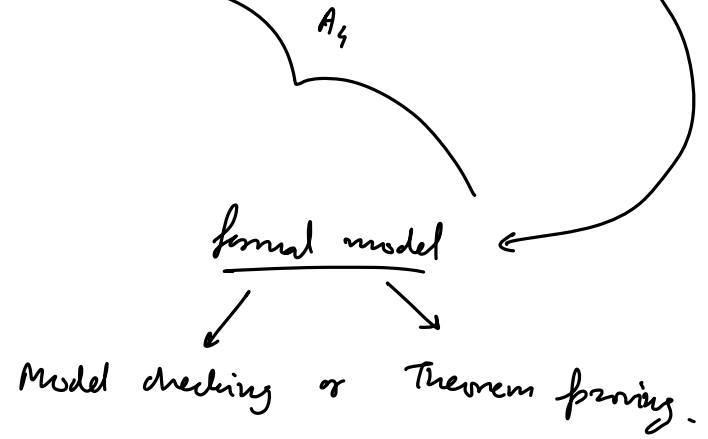
developing systems



formal development of software.

development
cycle





refactoring $\rightarrow$ migrating legacy code into

system description language $\rightarrow$ hybrid automata
 $\Rightarrow$ CTL

Specification style language,

$\rightarrow$ Z

$\rightarrow$ VDM

prog1 () {

3: $x_1 = x_2 + 5$;

4: $x_3 = (x_2 + x_1) * 3$;

5: $x_2 = x_1 * 9$;

return _ ;

}

Industrial case studies

→ TCAS II → English spec failed